

Hamza Ammar

Waterloo, Ontario — Email — LinkedIn — GitHub

EDUCATION

University of Waterloo

Bachelor of Software Engineering (BSE)

Elected Class Academic Representative

CERTIFICATIONS

- Machine Learning Specialization - *DeepLearning.ai, Stanford University* Aug 2025
- Algorithmic Toolbox - *UC San Diego* Jan 2025

EXPERIENCE

Shopify

Software Engineering Intern

Toronto, ON

May 2026 - Present

- Developing internal tools to support **merchant growth** across Shopify's platform

Dealish

Founding Engineer

Waterloo, ON

Dec 2025 - Present

- Built and launched **React Native (Expo)** location-based app for **real-time** food and drink discovery.
- Designed a secure **Supabase backend** with **role-based access control** to manage user profiles and deals.
- Integrated a **Google Sheets inventory** sync via OAuth for partner restaurants to **manage deals** in real-time

Kuzu Inc

Software Intern

Waterloo, ON

Jun 2024 - Aug 2024

- Built **operator printing** for KuzuDB's query planner, improving plan visualization and **cutting debugging time**
- Added metadata printing for **40+ operators**, **increasing clarity** of query plans used in performance benchmarking
- Improved Explorer design by refining node layouts and using **Chroma.js** pastel colours for a **cleaner interface**

MapFLOW

Data Manager

Waterloo, ON

Nov 2022 - Mar 2024

- Structured medical research into **JSON datasets** powering MapFLOW's app, trusted by **3,500+ pharmacists**
- Streamlined data access, reducing consultation time by **~15 minutes** and driving **10x pharmacy revenue**
- Built a **Python CLI** with **OpenAI's API** for data translation, expanding to **500+ French-speaking pharmacies**

PROJECTS

Horizon MCP

January 2026 - May 2026

- Deployed an **MCP academic assistant** on AWS serving course content, notes, and discussions to **50+ students**
- Built a **semantic retrieval pipeline** using **PDF embeddings and vector search** for quick notes access
- Integrated **D2L/Brightspace, Piazza and Crowdmark** with **asynchronous, fault-tolerant syncing**

Godseye

March 2026

- Built a **multi-agent** prediction market tool using **AI-driven simulations** on Polymarket bets
- Placed **2nd** in the Polymarket track at **YHack (Yale)**
- Designed **agent orchestration pipeline** where **specialized agents** debate and produce consensus trading signal

Deep Learning Chess Engine

November 2025

- Developed a **deep CNN-based chess engine** trained end-to-end on millions of grandmaster games
- Implemented a **residual convolutional architecture** with policy and value heads using **PyTorch**.
- Optimized training on **NVIDIA H100s** via efficient tensor operations, large batch sizes, and mixed precision

CourseConnect

Sep 2025 - Dec 2025

- Built a **Python + Playwright scraper** to collect and normalize **1,000+** courses into a structured dataset.
- Constructed a **prerequisite dependency graph** enabling SE students to **validate and plan degree paths**

SKILLS

- Languages:** Python, C++, C, Swift, JavaScript (React, Next.js, Node.js), SQL, Ruby
- Frameworks & Libraries:** TensorFlow, PyTorch, NumPy, Pandas, NetworkX, Playwright, Flask, Express.js
- Cloud & Tools:** AWS (EC2, S3), Firebase, Git/GitHub, Linux CLI, Docker, Graphite
- Technical Areas:** Full-stack Development, Machine Learning, Web Scraping, Algorithms and Data Structures